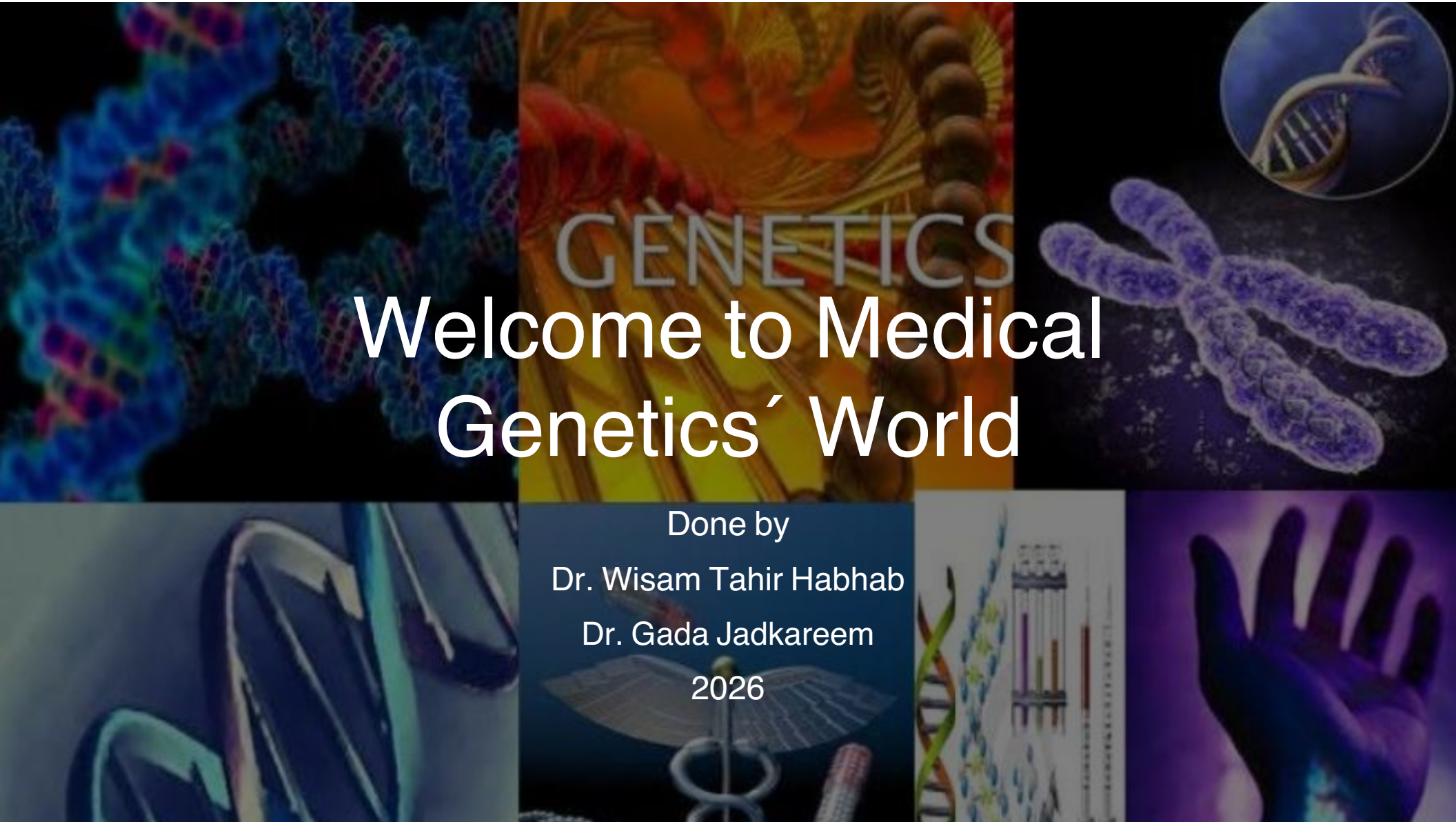




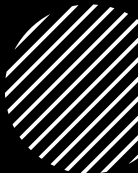
GENETICS

Welcome to Medical Genetics' World

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Why Study Medical Genetics?



Genetics helps explain a lot of things, like what makes you unique, why you look like other members of your family, why somebody looks different and why some diseases run in your family.



We study genetics because the DNA sequence is the beginning, the blueprint for our development and so many of the features that make us the unique individuals we are.



[Understanding Heredity](#)

Why Medical Genetics is Important?

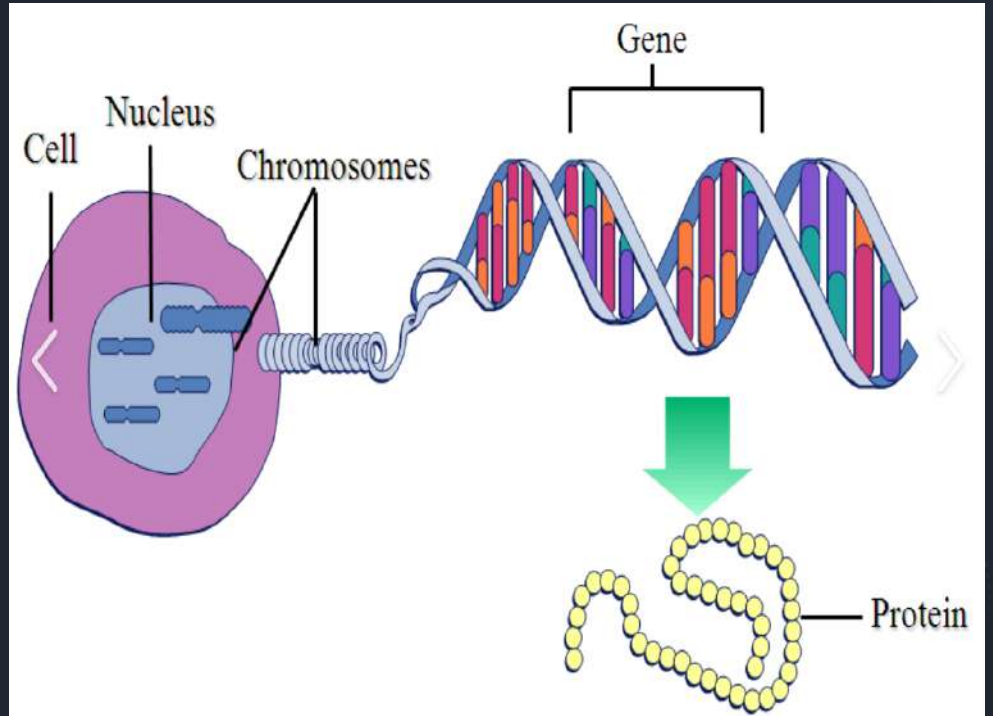
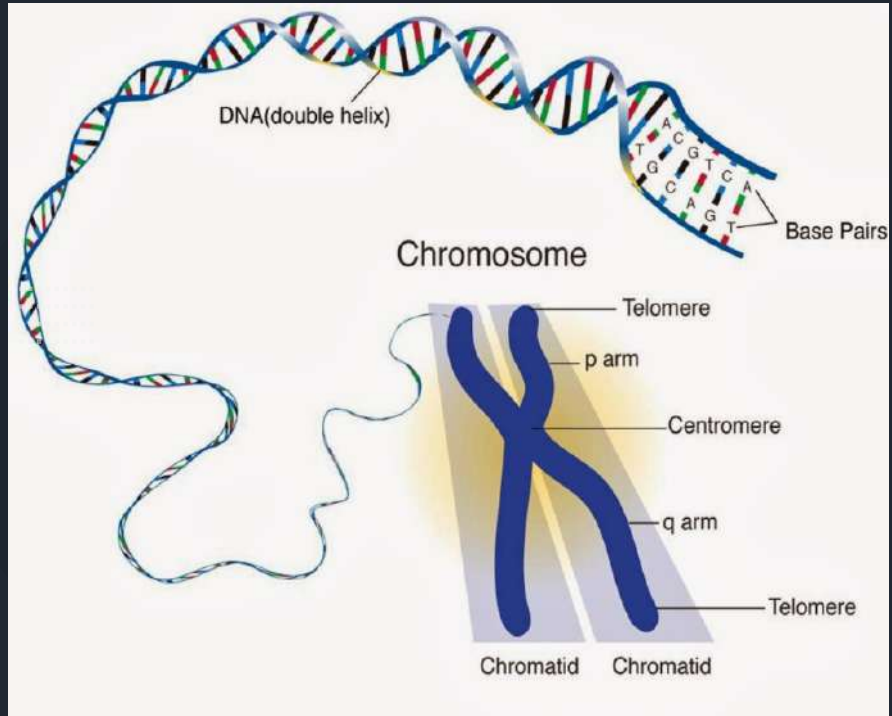
Medical genetics is crucial for several reasons:

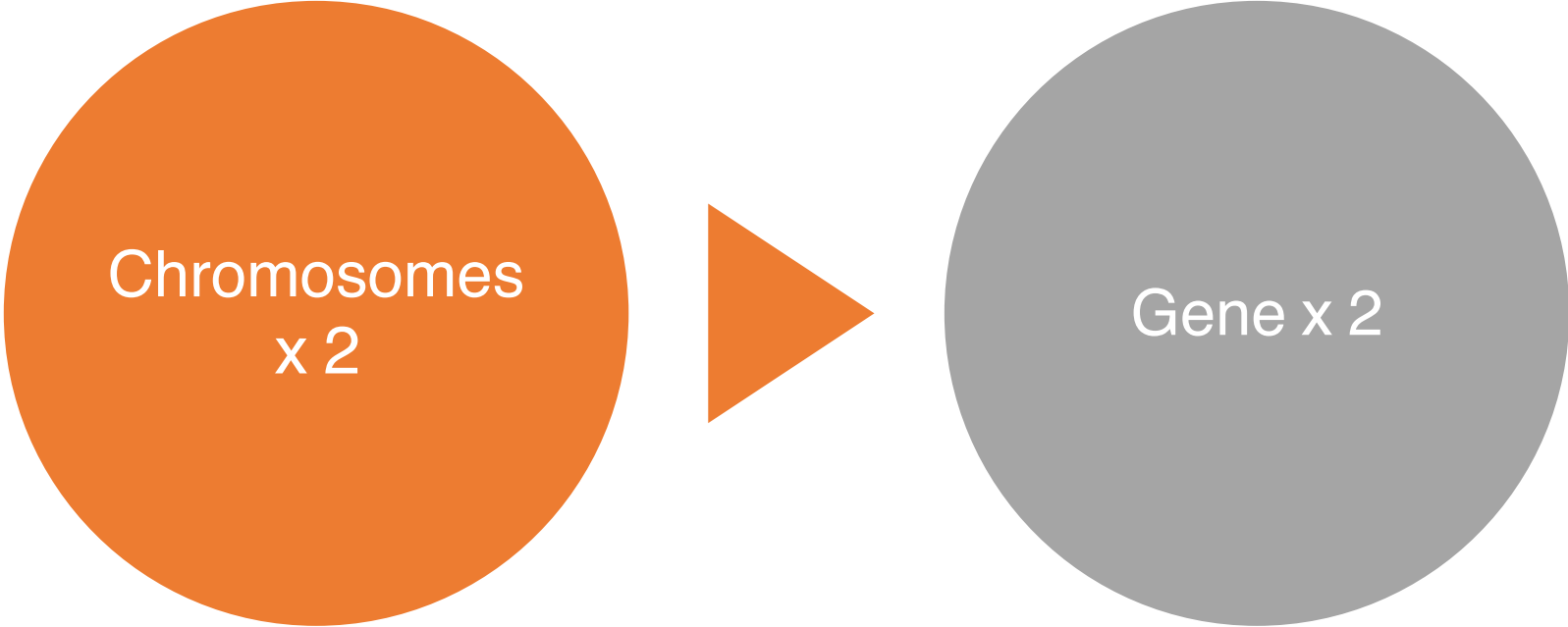
1. **Understanding Heredity:** How traits are passed down through generations, which is essential for diagnosing and treating genetic disorders.
2. **Personalized Medicine:** By tailoring treatments to an individual's genetic makeup, medical genetics can lead to more effective and safer therapies.
3. **Pharmacogenomics:** This field studies how an individual's genes affect their response to medications, allowing for more precise prescribing.
4. **Evolutionary Insights:** It provides insights into human evolution and the genetic differences between populations, aiding in the understanding of migration patterns and evolutionary history.

Medical genetics is a vital component of modern medicine, offering personalized care and improving the effectiveness of treatments.

- We are genes....







Chromosomes
x 2

The diagram consists of two large circles, one orange on the left and one gray on the right, connected by a right-pointing orange triangle. The orange circle contains the text 'Chromosomes x 2' and the gray circle contains the text 'Gene x 2'. The top of the image has a dark blue header bar.

Gene x 2

We will deal with the following branches in this module:

Basic Genetics (basic information in the field)

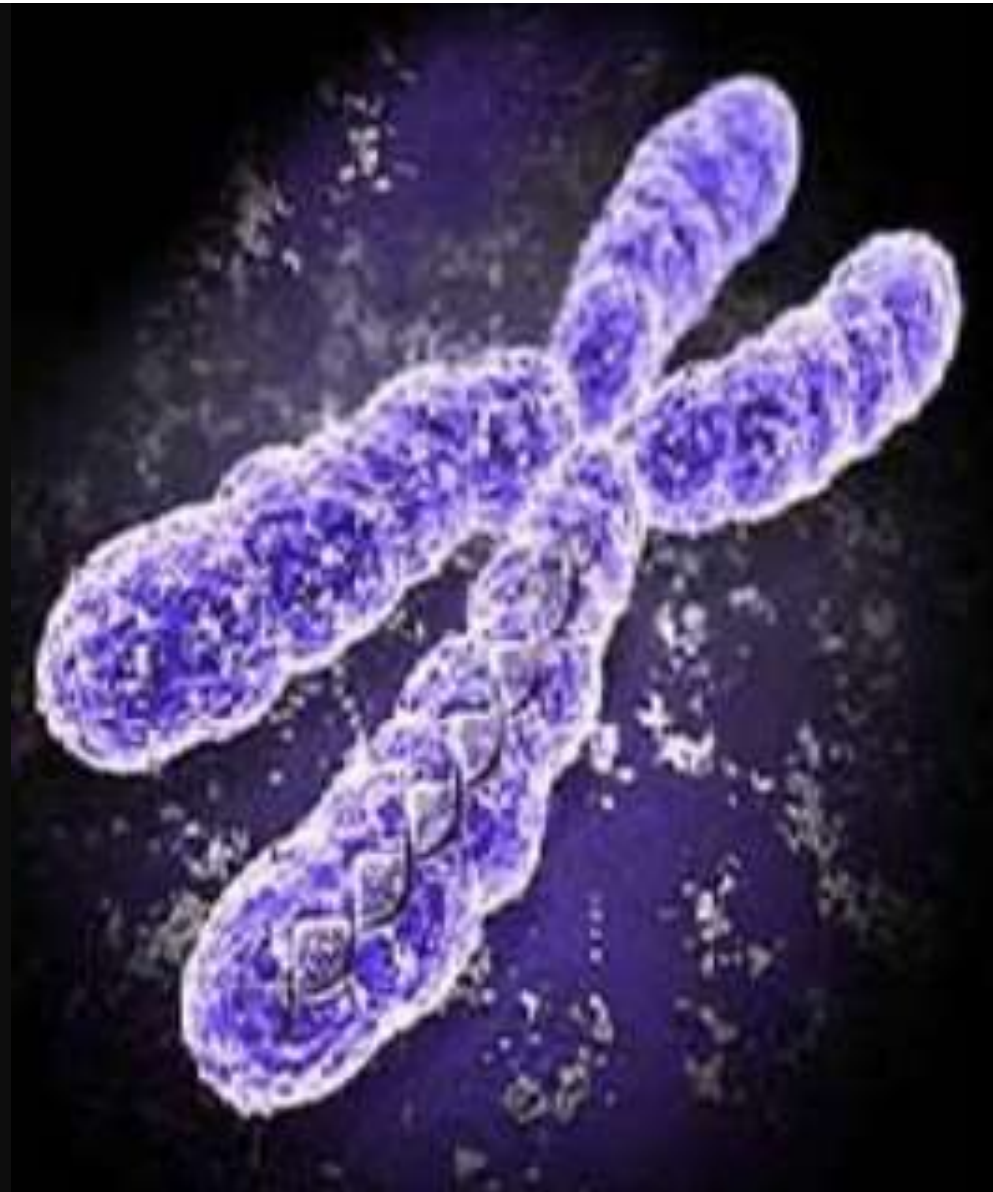
Cytogenetics (chromosomes)

Molecular Genetics (genes)

Clinical Genetics (syndromes)

What is Chromosome?

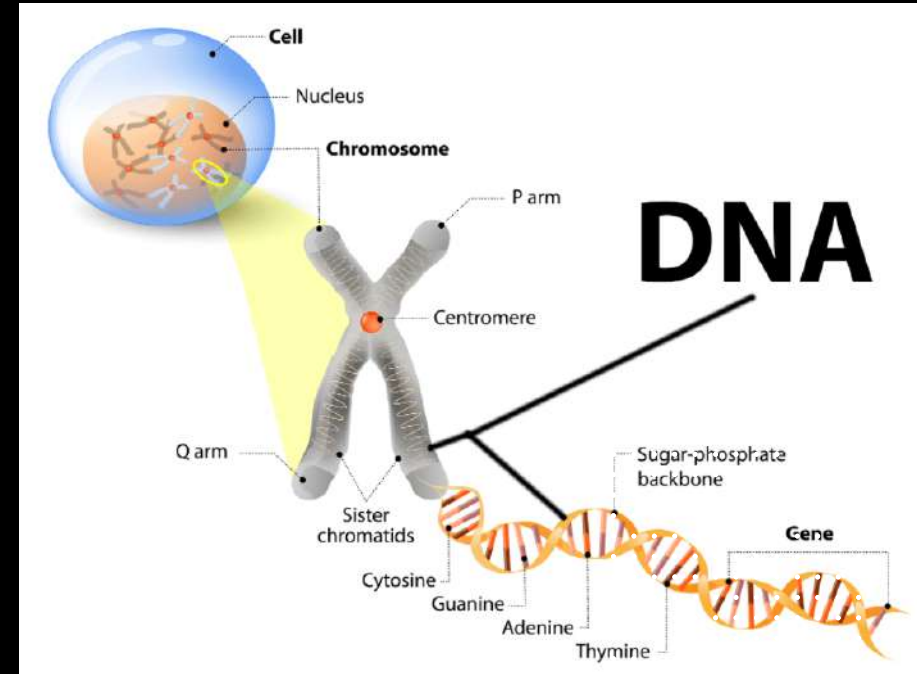
- A threadlike structure of nucleic acids and protein found in the nucleus of most living cells, carrying genetic information in the form of genes.
-

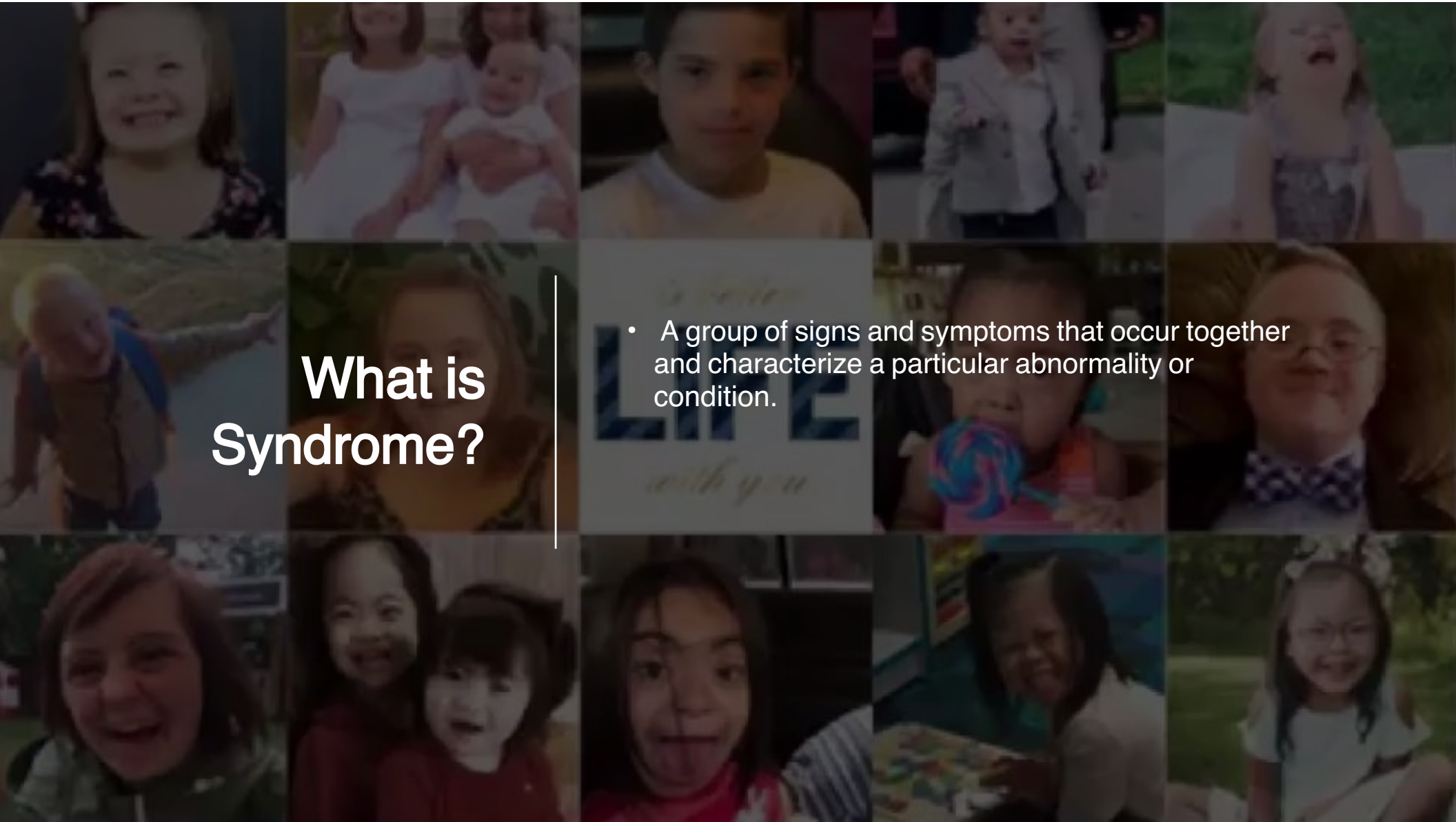


What is Gene and Gene Mutation?



- **Gene** is a unit of heredity which is transferred from a parent to offspring and is held to determine some characteristic of the offspring.
- **Gene** is a unit of hereditary information that occupies a fixed position (locus) on a chromosome. Genes achieve their effects by directing the synthesis of proteins. Genes are made up of promoter regions and alternating regions of introns (noncoding sequences) and exons (coding sequences).
- **Gene Variant** is a permanent alteration in the DNA sequence that makes up a gene, such that the sequence differs from what is found in most people.





What is Syndrome?

- A group of signs and symptoms that occur together and characterize a particular abnormality or condition.

General information about the course

(Please review the study guide thoroughly, as it contains detailed information about the course)

- Study Guide
- Exams
- Community service

من أشعار د. وسام حبيب

نحن أمة القرطاس والقلم

لا نقعدن عن الأحلام والهمم

مبتغانا العلا ومطلبنا القمم

نقرأ ونتعلم حتى بإذن الله نسود الأمم

Enjoy with
Genetics.....

Good Luck

